

Polymers

Monomers, repeat units, addition and condensation

Name _____ Class _____ Date _____

DATA

$n = \text{molar mass of polymer} \div \text{molar mass of repeat unit}$

Molar masses (g/mol): C 12.01 · H 1.008 · F 19.00 · C₂H₄ 28.05 · C₃H₆ 42.08 · C₂F₄ 100.02 ·
CO₂ 44.01 · H₂O 18.02 · nylon-6,6 repeat unit C₁₂H₂₂N₂O₂ 226.32

Show units on every quantity in every step.

A. Understanding

- Define (a) monomer (b) repeat unit (c) addition polymerization (d) condensation polymerization.

B. Repeat units and monomers

- Draw the repeat unit of the polymer formed from: (a) propene, CH₂=CHCH₃ (b) chloroethene, CH₂=CHCl (c) tetrafluoroethene, CF₂=CF₂ (d) phenylethene (styrene), CH₂=CHC₆H₅

3. Give the monomer of each polymer: (a) $-\text{[CH}_2\text{-CH(CN)]}-$ (b) $-\text{[CH}_2\text{-CH(CH}_3\text{)]}-$ (c) $-\text{[CH}_2\text{-CCl}_2\text{]}-$

4. Classify each as an addition or a condensation polymer: poly(ethene), nylon-6,6, PET (a polyester), PVC, a protein, PTFE.

5. Nylon-6,6 is made from hexanedioic acid, $\text{HOOC(CH}_2\text{)}_4\text{COOH}$, and hexane-1,6-diamine, $\text{H}_2\text{N(CH}_2\text{)}_6\text{NH}_2$. (a) Name the type of link formed and draw it. (b) What small molecule is released?

C. Calculations

6. (a) A PTFE sample has an average molar mass of 3.00×10^5 g/mol. Find the average number of repeat units. (b) A poly(propene) chain has 3.00×10^3 repeat units. Find its molar mass.

7. Calculate the percentage by mass of fluorine in PTFE.

8. Forming each repeat unit of nylon-6,6 releases 2 molecules of water. What mass of water is released when 1.00 kg of nylon-6,6 is made?

D. Challenge

9. (a) Write a balanced equation for the complete combustion of poly(ethene), using one repeat unit, C_2H_4 . (b) What mass of CO_2 forms when 1.00 kg of poly(ethene) waste is burned? (c) Give one advantage and two disadvantages of burning plastic waste compared with recycling it.